Quantitative Measurement of Temporal Concordance and Stability in Cryptoasset Taxonomies

A Multi-View Latent Framework Incorporating Physical Infrastructure Signals

Introduction, literature gap, research questions and objectives

Cryptoasset markets are usually analysed through volume, liquidity and TVL (total value locked) as a unified market. However, a token or coin alone is only a single representation of the underlying asset. The same asset may function as an exchange instrument, collateral object, decentralised software project or an attention magnet. Existing taxonomy labels are useful, but they are not ground truth. The central gap is that cryptoasset taxonomies are rarely tested as empirical structures across on-chain, market, developer, attention (speculation), physical-infrastructure and remote-sensing views. This research tests whether different taxonomies reveal distinct latent structures and whether, and if so when those structure remain stable across views and time.

Current cryptoasset and distributed-ledger taxonomies help organise assets but they do not answer the key quantitative question whether those labels remain visible, stable and comparable when measured through market, on-chain, developer, attention, infrastructure and remote-sensing views [1-3]. Crypto-finance and DeFi research shows that returns, liquidity, token adoption and protocol activity contain measurable structure, but are usually studied separately rather than part of a taxonomy stability framework [4-7]. Multi-view representations, dimensionality reduction and partition-comparison methods provide the tools to compare representations, while contagion literature shows why apparent convergence during outlier events must be interpreted carefully [8-13].

This proposal extends the logic of my MSc project [14], which compares representations using concordance, stability and spatial diagnostics. This PhD keeps the diagnostic logic but changes the domain of the shared raster grid into a multi asset comparison, spatial instability becomes asset overlap and the diagnostic framework becomes a reproducible quantitative-finance evidence chain.

The main research question is: How can multi-view data be used to reconstruct, compare and track the latent structure of cryptoasset taxonomies over time, and what do changes in taxonomy concordance reveal about statistical anomalies, risk monitoring and predictive value? The aim is to reconstruct cryptoassets as multi-view latent objects, aggregate them into taxonomy-level structures, and test whether those structures agree, diverge, drift or become unstable. The project is alpha-aware, not alpha-claiming. With predictions and back-testing used only to test information beyond price and the random baselines not to prove a trading edge.

Research Questions and Objectives

RQ	Research Question	Objectives
RQ1	Can individual cryptoassets be represented as multi-view latent objects using on-chain, market, developer and physical infrastructure data?	O1: Build a multi-view cryptoasset dataset. O2: Create asset-level latent representations. O3: Test physical infrastructure signal value.
RQ2	To what extent do cryptoasset taxonomies reveal measurable concordance and divergence across multiple data views?	O4: Build taxonomy-level metrics. O5: Measure taxonomy concordance and divergence.
RQ3	How stable are taxonomy relationships over time, and when do they converge, diverge or reveal statistical anomalies?	O6: Track taxonomy drift and anomalies over time. O5
RQ4	Do latent structures, taxonomy harmony and divergence signals improve prediction and risk monitoring beyond price data alone?	O7: Test prediction and risk-monitoring value beyond price data. O2, O5, O6

Background: literature review, MSc link and contribution

This research crosses between taxonomy, crypto-finance, multi-view representation, contagion and infrastructure research. Existing cryptoasset taxonomies classify assets but do not fully test whether those classifications remain visible across on-chain, market, developer, attention, infrastructure and remote-sensing views [1-3].

Crypto-finance and DeFi studies show that liquidity, adoption and protocol activity contain measurable structure, but these signals are usually measured separately rather than evidence of taxonomy stability [4-7]. Multi-view representations PCA, UMAP and partition comparison methods provide the fundamental tools for comparing representations, as implemented in my MSc project [14]. While contagion research shows how divergence happens during stress which should be interpreted cautiously [8-13].

The thesis does not aim to create a final universal cryptoasset taxonomy or claim alpha. It tests whether taxonomy labels are coherent across views, stable through time and useful for risk monitoring beyond price-only and random baselines. The preliminary taxonomy table below defines the categories being tested, not assumed.

The core contribution is a reproducible framework for testing whether taxonomy labels are consistent enough to support risk monitoring, anomoly detection and empirical comparison rather than truth of taxonomies. The research does not aim to produce a universal cryptoasset taxonomy assignment nor does it aim to claim alpha, this matters as downstream risk and prediction results are not always stable. Literature Gap Table (Appendix 1) & Preliminary Taxonomy Benchmarks (Appendix 2).

The methodology develops and aggregates a multi-view dataset which tests taxonomies of cryptoassets across on-chain, market, developer, attention, physical-infrastructure and remote-sensing layers. With the data standardised and converted into latent embeddings using dimensionality reduction [8-10]. Clustering and partition comparison metrics, test whether taxonomy structures agree or diverge across views [9-11]. Rolling windows and Kalman-style turning point detection tracks temporal change, drift, convergence and anomalies over time. Finally, price-only, random baselines and walk-forward validation will be used for alpha aware validation rather than alpha claiming. Data Sources (Appendix 3).

Papers produced: Objectives, Requirements and deliverables linked

The thesis will be six chapters, with chapter 1 introducing the thesis and chapter 6 concluding it. Chapter 2 explicitly connects the PhD to the MSc framework. Chapters 3, 4 and 5 which will be empirical publishable papers, which builds upon the methodology.

CH	Title	Role	RQ
1	Introduction to latent structure reconstruction in cryptoasset markets	Problem, motivation, research questions and contribution.	RQ1-RQ4
2	Literature review, methodology and MSc diagnostic-framework link	Reviews crypto taxonomies, multi-view representations, quantitative finance, infrastructure signals and explains how the MSc concordance and stability logic expands into the PhD	RQ1-RQ4
3	Revealing Latent Cryptoasset Structures: A Multi-View Framework Incorporating Physical Infrastructure	Builds the individual cryptoasset multi-view latent objects and explore physical infrastructure as a layer	RQ1
4	Measuring Multi-View Concordance and Divergence Across Cryptoasset Taxonomies	Aggregates cryptoassets into taxonomies and measures concordance, divergence and overlap	RQ2
5	Beyond Price: Temporal Drift Across Cryptoasset Taxonomies	Tracks stability, drift, convergence, statistical anomalies and alpha-aware validation	RQ3, RQ4
6	Conclusion, contribution, limitations and use cases	Consolidates methodology, quantitative-finance and reproducibility contributions to answer the RQs	RQ1-RQ4

Combined

RQ	Requirement	Evidence	Deliverable
RQ1	R1: Build multi-view cryptoasset dataset. R2: Create asset-level latent representations. R3: Test infrastructure signal value. R8: Ensure reproducibility and auditability.	E1: Coverage, identifiers, duplicates, source audit. E2: Embeddings, profiles, stability, variance, divergence. E3: Coverage, contribution score, model comparison. E8: Code, configs, SQL, logs, archived outputs.	D1: Multi-view cryptoasset dataset. D2: Data quality audit and report. D3: Individual cryptoasset embeddings, profiles and outputs. D4: Physical infrastructure feature set and contribution report., D9: Figures, tables, and chapter-ready empirical outputs., D10: Reproducible git code archive, config files, logs, and final output package.
RQ2	R4: Build cryptoasset taxonomies. R5: Measure taxonomy agreement and divergence. R8	E4: Rules, categories, balance, sensitivity. E5: Concordance, overlap, harmony, divergence indices. E8	D5: Taxonomy rules, category definitions and metrics. D6: Taxonomy concordance, harmony, divergence and overlap outputs. D9, D10
RQ3	R6: Track taxonomy change over time. R5, R8	E6: Rolling estimates, drift, convergence, anomalies. E5, E8	D7: Rolling window stability, drift, convergence and anomaly outputs. D6, D9, D10
RQ4	R7: Validate prediction and risk value. R2, R5, R6, R8	E7: Random and price-only benchmarks, risk metrics, costs. E2, E5, E6, E8	D8: Prediction and risk-monitoring validation outputs against price data. D3, D6, D7, D9, D10

Possible implementations and why they matter

There are four possible ways this framework could be implemented, they all matter as they turn taxonomy labels into testable evidence rather than fixed assumptions. These labels are used in risk attribution, portfolio grouping and research, but currently there is still no clear auditable framework to test whether those labels remain empirically stable.

- **Taxonomy audit package:** Tests category balance, concordance and divergence before labels are used in research or risk models. Foundation of paper 2 and link to taxonomy literature[1-3].
- **Risk-monitoring framework:** Track, harmony, divergence, drift and anomaly signals over time. Foundation of paper 3 and links to cross dependent evidence [12,13].
- **Alpha-aware validation:** Compare latent structure signals with price only and random benchmarks which links to R7, R8 and crypto finance evidence [4-7].
- **Infrastructure-exposure monitor:** Report validator, node, energy, data-centre and remote-sensing indicators where measurable linking to R3, R6, R8.

Feasibility, risk and Gantt chart

The proposed PhD Timescale (Appendix 4) is feasible as it aligns with the staged order of the project, with the evidence chain, scope, data, reconstruction, taxonomy comparison, temporal analysis then thesis integration. The main risks and mitigations:

- **Uneven infrastructure data:** Only use infrastructure and remote-sensing data when relevant to asset.
- **Taxonomy imbalance:** Define and archive taxonomy versions and test alternative definitions (hyper-parameter testing).
- **Method complexity:** Experiment with bootstrapped features and methods to find the core concordance, stability and drift over time, ensuring robustness.
- **Over promising:** Treat as alpha-aware model not alpha-claiming, show temporal divergence metrics

Conclusion and expected contribution

This PhD proposes a reproducible quantitative framework for measuring temporal concordance and stability in cryptoasset taxonomies. The contribution is important because taxonomy labels shape how cryptoasset markets are interpreted, sampled, risk-managed and validated. If taxonomy labels are unstable across views, then risk-monitoring and prediction results built on those labels can be misleading. If latent structures remain stable across views and time, then they provide stronger evidence for category-level interpretation. If they drift or diverge, that drift is itself useful evidence. While many methods will be tested, the purpose of this research is to reveal whether there are latent structures across the data views.

The expected contribution is threefold. Methodologically, the thesis extends the MSc diagnostic logic from spatial clustering to cryptoasset latent-structure reconstruction. Empirically, it produces auditable datasets, taxonomy indices, harmony/divergence measures, boundary reports and rolling anomaly outputs. Practically, it can support taxonomy design, risk monitoring, infrastructure exposure analysis and alpha-aware validation. The thesis should therefore be judged on whether it makes cryptoasset structure observable, comparable and auditable across views and through time, not on whether it claims to produce alpha.

References

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Appendix 1

Paper	Taxonomies/ Classification	Multi-view/ Latent Structures	Concordance/ Divergence	Temporal Stability/ Anomalies	Alpha-aware Validation	Infrastructure
Tasca et al. (2019)	✓					
Ankenbrand et al. (2020)	✓					
Ballandies et al. (2022)	✓					
Liu et al. (2021)					✓	
Makarov et al. (2020)				✓	✓	
Zhao et al. (2017)		✓				
Jolliffe et al. (2016)		✓				
McInnes et al. (2018)		✓				
Robert et al. (2021)			✓			
Forbes & Rigobon (2002)				✓		
Galati et al. (2024)				✓	✓	
Stoll et al. (2019)						✓
Kian (2026)		✓	✓	✓		✓
This PhD	✓	✓	✓	✓	✓	✓

Appendix 2

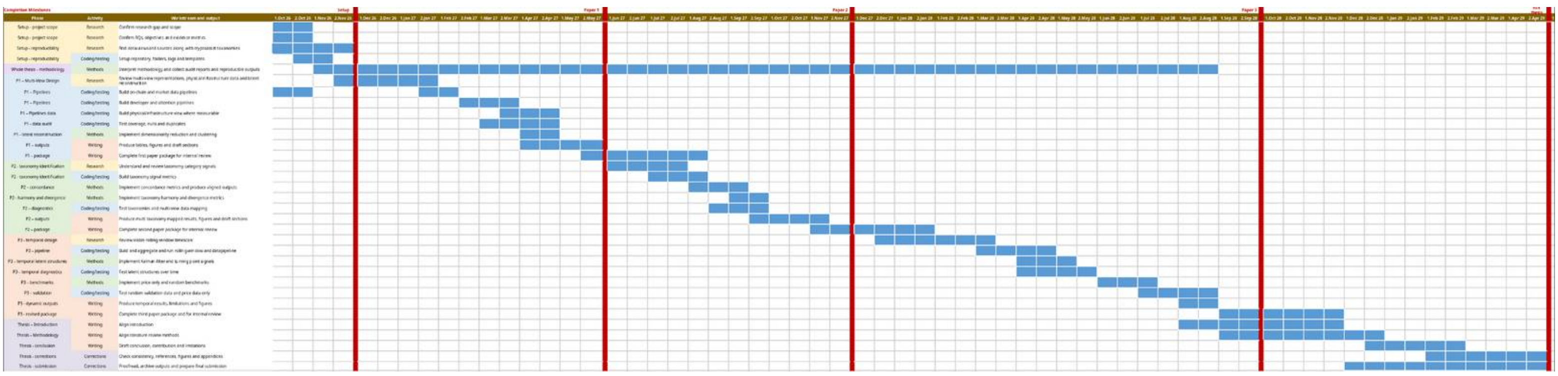
Layer	Examples	Purpose
Monetary assets	BTC-like projects	Macro benchmark
Core networks	Layer 0s, Layer 1s, Layer 2s	Infrastructure and scaling
Exchange assets	Exchange tokens	Centralised-market structure
DeFi protocols	AMMs, lending, derivatives	Liquidity, credit and leverage
Middleware	Oracles, bridges	Cross-protocol connectivity
Physical infrastructure	DePIN, nodes, validators	Real-world infrastructure exposure
Privacy assets	Monero-like projects	Regulatory and behavioural layer
Consumer/speculation assets	Gaming, NFTs, memecoins, AI-agent tokens	Application demand and retail narratives

Appendix 3

Layer	Data	Source	Meaning
On-chain	State, activity, usage, fees, TVL, bridges	RPC, BigQuery, DefiLlama	Blockchain activity
Market	Price, volatility, liquidity, volume, market cap	CoinGecko, CCData, Kaiko	Market behaviour
Developer	Commits, contributors, releases, issues, PRs	GitHub API	Builder activity
Attention	Search, news, social media, narratives	Reddit, X, CourtListener	Public/media attention
Physical infrastructure	Energy, carbon, renewables, node/validator/storage geography	CBECI, EIA, ENTSO-E, Electricity Maps	Real-world infrastructure exposure
Remote sensing	Heat, water proximity, expansion signals	VIIRS, Sentinel, MODIS, Landsat	Satellite evidence of physical activity
Model outputs	Embeddings, clusters, concordance, divergence, drift, anomalies	Model pipeline	Taxonomy structure and change

Appendix 4

Phase	Activity	Work-stream and output
Setup - project scope	Research	Confirm research gap and scope
Setup - project scope	Research	Confirm RQs, objectives and evidence metrics
Setup - reproducibility	Research	Find data-views and sources along with cryptoasset taxonomies
Setup - reproducibility	Coding/testing	Setup repository, folders, logs and templates
Whole thesis - methodology	Methods	Interpret methodology and collect audit reports and reproducible outputs
P1 – Multi-View Design	Research	Review multi-view representations, physical infrastructure data and latent reconstruction
P1 – Pipelines	Coding/testing	Build on-chain and market data pipelines
P1 – Pipelines	Coding/testing	Build developer and attention pipelines
P1 – Pipelines data	Coding/testing	Build physical/infrastructure view where measurable
P1 - data audit	Coding/testing	Test coverage, nulls and duplicates
P1 - latent reconstruction	Methods	Implement dimensionality reduction and clustering
P1 – outputs	Writing	Produce tables, figures and draft sections
P1 – package	Writing	Complete first paper package for internal review
P2 - taxonomy identification	Research	Understand and review taxonomy category signals
P2 - taxonomy identification	Coding/testing	Build taxonomy signal metrics
P2 – concordance	Methods	Implement concordance metrics and produce aligned outputs
P2 - harmony and divergence	Methods	Implement taxonomy harmony and divergence metrics
P2 – diagnostics	Coding/testing	Test taxonomies and multi-view data mapping
P2 – outputs	Writing	Produce multi taxonomy mapped results, figures and draft sections
P2 – package	Writing	Complete second paper package for internal review
P3 - temporal design	Research	Review viable rolling window timescale
P3 – pipeline	Coding/testing	Build and aggregate and run rolling-window and data pipeline
P3 – temporal latent structures	Methods	Implement Kalman-filter and turning point signals
P3 – temporal diagnostics	Coding/testing	Test latent structures over time
P3 – benchmarks	Methods	Implement price-only and random benchmarks
P3 – validation	Coding/testing	Test random validation data and price data only
P3 - dynamic outputs	Writing	Produce temporal results, limitations and figures
P3 - revised package	Writing	Complete third paper package and for internal review
Thesis – Introduction	Writing	Align introduction
Thesis – Methodology	Writing	Align literature review methods
Thesis - conclusion	Writing	Draft conclusion, contribution and limitations
Thesis - corrections	Corrections	Check consistency, references, figures and appendices
Thesis - submission	Corrections	Proofread, archive outputs and prepare final submission



Red lines indicate completion of (Nov 26), paper 1 (May 27), paper 2 (Nov 27), paper 3 (Sep 28) and finish full thesis (Apr 29). The structure is to give an idea of the viability of the project across October 2026 to May 2029.